

NCERT Class 12- Human Geography gist – ORACLE IAS

Chapter 1- Human Geography (nature and scope)

Nature of Human Geography

- Human Geography studies the inter-relationship between the physical environment and socio- cultural environment created by human beings through mutual interaction with each other.
- The elements like villages, cities, road-rail networks, etc and all other elements of material culture have been created by human beings using the resources provided by the physical environment. Thus, In the saying of Ellen Semple “Human geography is a study of changing relationship between unresting man and unstable earth”.

Naturalisation of Humans and Humanisation of Nature

- Humans interact with their physical environment with the help of technology. This indicates the level of cultural development.
- The interaction of primitive societies with the physical environment is termed as environmental determinism which is naturalisation of humans.
- With the development of technology, humans began to modify nature and created cultural landscape. This is called possibilism or humanisation of nature.
- A middle path of neo determinism was introduced by Griffith Taylor which means that neither is there a situation of absolute necessity (environmental determinism) nor is there a condition of absolute freedom (possibilism).

History of Human Geography

- The emergence of human geography started with the interaction, adaptation, adjustment and modification between the humans and the environment.
- Before the age of discovery, there was very little interaction between different societies but in the late 15th Century information about the unknown societies were made available now. Exploration by travellers expanded the area of human geography and interacted with different societies.
- With this, new approaches can across like welfare or humanistic school of thought, radical school of thought and behavioural school of thought.

Field and Sub-fields of Human Geography

- Human geography is inter-disciplinary in nature and develops vast linkages with other sister disciplines in social sciences.

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- The fields and sub-fields of human geography explains every aspect of all elements of human life on the surface of the earth.

Human Geography and Sister Disciplines of Social Sciences

Fields of Human Geography	Sub-Fields	Interface with Sister Disciplines of Social Sciences
Social Geography	—	Social Sciences-Sociology
	Behavioural Geography	Psychology
	Geography of Social Well-being	Welfare Economics
	Geography of Leisure	Sociology
	Cultural Geography	Anthropology
	Gender Geography	Sociology, Anthropology, Women's Studies
	Historical Geography	History
	Medical Geography	Epidemiology
Urban Geography	—	Urban Studies and Planning
Political Geography	—	Political Science
	Electoral Geography	Psephology
	Military Geography	Military Science
Population Geography	—	Demography
Settlement Geography	—	Urban/Rural Planning
Economic Geography	—	Economics
	Geography of Resources	Resource Economics
	Geography of Agriculture	Agricultural Sciences
	Geography of Industries	Industrial Economics
	Geography of Marketing	Business Studies, Economics, Commerce
	Geography of Tourism	Tourism and Travel Management
	Geography of International Trade	International Trade

Chapter 2 - The World Population

Patterns of Population Distribution

- Population distribution means arrangement distribution of people over the earth's surface. Population is not evenly distributed as 90 percent of the world's population lives in about 10 percent of its land area.

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- The 10 most populous countries of the world contribute about 60 per cent of the world's population. Out of these 10 countries, 6 are located in Asia.

Density of Population

- This means the ratio between the number of people to the size of the land. It is usually measured in persons per sq km density of population/area. Some areas are densely populated like North-Eastern USA, North-Western Europe, South, South-West and East Asia.
- Some areas are sparsely populated like near the polar areas and high rainfall zones near the equator while some areas have medium density like Western China, Southern India, Norway, Sweden, etc.

Factors Influencing Population Distribution

The population distribution is influenced by three factors i.e., geographical factors, economic factors and social and cultural factors.

Geographical Factors

Environmental or natural factors such as landforms, fertile soil, suitable climate for cultivation and availability of adequate source of fresh water are the geographical factors that affect the population distribution. Some geographical factors are:

Land Forms Flat Plains and gentle slopes are preferred by people, because these are favorable for the production of crops and to build roads and industries.

Climate Area with less seasonal variation attract more people.

Soil Area which have fertile loamy soil have more people living on them as these can support intensive agriculture.

Water People prefer to live in areas where fresh water is easily available. Because, it is the most important factor for life.

Economic Factors

Places having employment opportunities like 1 mineral rich areas, industrial units and urban centres have high concentration of population. Some economic factors are:

Industrialisation Industries provide job opportunities and attract large numbers of ' people.

Minerals Minerals deposits attract industries 1 mining and industrial activities generate employment.

Urbanisation Good civic amenities and the attention of city life draw people to the cities.

Social and Cultural Factors

Places having religious importance and cultural significance are also very densely populated areas.

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Population Growth

This refers to the change in number of inhabitants of a territory during a specific period of time. When change in population is expressed in percentage, then it is called Growth Rate of Population. When there is an increase in population by taking the difference between births and deaths, then it is called Natural Growth of Population. There is also Positive Growth of Population which happens when birth rate is more than death rate and Negative Growth of Population when birth rate is lower than death rate.

Components of Population Change

There are three components of population change i.e., births, deaths and migration.

Crude Birth Rate [CBR]

Number of births in a year per thousand of population is expressed as Crude Birth Rate (CBR). It is calculated as:

$$CBR = \frac{B_i}{P} \times 100$$

Here, B_i = live Births during the year; P = Mid year population of the area.

Crude Death Rate (CDR)

Number of deaths in a year per thousand of population is expressed as Crude Death Rate (CDR). It is calculated as:

$$CDR = \frac{D}{P} \times 100$$

Here, D = Number of Deaths; P = Estimated mid-year population of that year.

Migration

It is movement of people across region on permanent, temporary or seasonal basis. The place they move is called place of origin and the place they move to is called place of destination.

Push and Pull Factors of Migration

The Push factors make the place of origin seem less attractive for reasons like unemployment, poor living conditions, political turmoil, unpleasant climate, natural disasters, epidemics and socio-economic backwardness.

The Pull factors make the place of destination seem more attractive than the place of origin for reasons like better job opportunities and living condition ease and stability, security of life and property and pleasant climate.

Trends in Population Growth

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- Trends show that initially growth of population was very slow but after the improvement in Science and Technology, there had been tremendous growth in population which is called population explosion.
- About 8000 to 12000 years ago world population was 8 million and now it has reached to 7 billion.
- In every 12 years, 1 billion people are added. Increased agriculture and industrial production, inoculation against epidemics, improvement in medical facilities have reduced death rates.

Doubling Time of World Population

- Developed countries are taking more time to double their population as compared to developing countries.
- Oman, Saudi Arabia, Somalia, Liberia, Yemen have high population growth rates while Latvia, Estonia, Russia, Germany, etc have low growth rates.

Spatial Pattern of Population Change

- The world population growth rate is 1.4%, it is highest in Africa i.e. 2.6% and lowest in Europe i.e. 0.0% means neither grow nor decline.
- So even when a small annual rate is applied to very large population, it will lead to a large population change. There is negative correlation between economic development and population growth.

Impact of Population Change

High increase in population leads to problems like depletion of natural resources, unemployment and scarcity. Decline in population indicates that resources are insufficient to maintain a population.

Demographic Transition Theory

This theory studies the changes in the population of a region as it moves from high births and high deaths to low births and low deaths. This happens when a society progresses from rural agrarian and illiterate to urban, industrial and literate.

There are three-staged model of Demographic Transition Theory. They are:

First Stage

- This stage is marked by high fertility high mortality rate because people reproduce more to compensate for the deaths due to epidemics and variable food supply.

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- People are poor, illiterate and mostly engaged in agriculture. Life expectancy is low and population growth is slow.

Second Stage

- Level of technology increases and other facilities like medical, health, sanitation improves due to which the death rate reduces.
- But the fertility rate and birth rate remains high due to which there is huge rise in population. Population expands rapidly as there is wide gap between birth and death rate.

Third Stage

- The birth and death rate both reduces and the population moves towards stability.
- People become literate, urbanised and control the size of the family. There is good judicious use of technology also.

Population Control Measures

- Family planning is the spacing and preventing the birth of children. Thomas Malthus theory (1793) states that the number of people would grow faster than the food supply thus leading to famine, diseases and war.
- Therefore, it is essential to control the population. This is undertaken through measures like awareness for family planning, free availability of contraceptives, tax disincentives and active propaganda.

Chapter 3 - Population Composition

Sex Composition

- Sex composition refers to the number of men and women in a country. It is taken out as a sex ratio which is the ratio between the number of men and women. It is calculated by the formula:

$$\frac{\text{Male Population}}{\text{Female Population}} \times 100 \text{ or } \frac{\text{Female Population}}{\text{Male Population}} \times 100 \text{ (In India this formula is applied)}$$

- Sex ratio in the world is 990 females per 1000 males'with highest in Latvia (1187 females per 1000 males) and lowest in UAE (468 females per 1000 males).
- The sex ratio is unfavourable in 72 countries of the world as there is gender discrimination leading to female foeticide, female infanticide as well as low economic status of women.

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- In general, Asian countries like China, India have low sex ratio while European countries have higher sex ratio.

Age Structure

The age structure of a population refers to the number of people of different age groups. It is divided into three groups i.e. children (0-14 years), adult or working population (15-59 years) and old or ageing population (60 years and above).

Age-sex Pyramid

The age-sex pyramid refers to the number of females and males in different age groups. The characteristic of a population is shown by population pyramid. The pyramid is interpreted in the following ways:

Expanding Population A triangular pyramid with a wide base due to high birth rate is mostly in less developed countries like Nigeria, Mexico and Bangladesh.

Constant Population A bell shaped pyramid and tapered towards the top shows almost equal birth and death rates like in Australia.

Declining Population A pyramid with narrow base and top with a wider central part shows low birth and death rates. This is in developed countries like Japan.

Rural Urban Composition

- The division of rural and urban areas based on residence, indicates the pattern of occupations and social conditions. People are predominantly engaged in primary activities in rural areas while in urban areas majority of working population is engaged in non-primary activities.
- In developed countries like Canada, West European countries, females outnumber males in urban areas while in developing countries like Nepal and Pakistan, it is vice versa.

Literacy

- Level of literate population of a country indicates the standard of living, social status of females, availability of educational facilities.
- In India literacy rate is the percentage of population above 7 years who are able to read, write and do arithmetic calculations.

Occupational Structure

- This is proportional distribution of working population (age group 15 to 59) in various occupations. Agriculture, forestry, fishing and mining are classified under primary activities.
- Manufacturing as secondary activity, transport, communication and other services as tertiary and jobs related to research and developing ideas are quaternary activities. In

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developed economics more number of people are engaged in secondary, tertiary and quaternary sector than primary.

Chapter 4 - Human Development

Growth and Development

- Growth and development refers to changes over a period of time but the difference between growth and development is that growth is quantitative but development is qualitative. Therefore, development is always positive.
- Development cannot take place unless there is an increment or addition to the existing conditions, but growth can be positive, negative or neutral, positive growth does not always led to development.
- Development occurs when there is a positive change in qualities. Earlier economic growth and development of country was seen as one but now they are studied separately.

Human Development

- This concept was introduced by **Dr Mahbub-ul-Haq** in 1990. Human development is described as that which enlarges people's choices and improves their lives, thus creating conditions where people can lead meaningful lives.
- This means that people develop talent, participate in society and be free to achieve their goals. According to **Prof. Amartya Sen**, the main objective of development is an increase in freedom. Freedom to make choices brings development, and social and political institutions play a major role in increasing freedom.
- Building people capabilities in health, education and equal access to resources increases freedom and enlarges choices.

Four Pillars of Human Development

The idea of human development is supported by the concepts of the following four pillars of development:

Equity It refers to equal access to opportunities available to everybody irrespective to their gender, race, income and caste.

Sustainability It means continuity in the availability of opportunities so that future generations can even use the present resources.

Productivity It means human labour which must be enriched by building capabilities in people.

Empowerment It means having the power to make choices which is possible by increasing freedom and capability.

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Approaches to Human Development

There are four ways of looking at or approach the problems of human development. Some of the important approaches are:

Income Approach This approach links the development to incomes as it believes that income determines the level of freedom that one enjoys.

Welfare Approach Under this approach, government is responsible for providing basic facilities like health, education and amenities to people.

Basic Needs Approach In this approach, emphasis is on providing six basic needs i.e. health, education, food, water supply, " sanitation and housing.

Capability Approach This approach is associated with Prof. Amartya Sen and aims to build human capabilities in health, education and-access to resources in order to increase human development.

Measuring Human Development

Following are the ways of measuring human development

Human Development Index

- Human development is measured through Human Development Index (HDI) which ranks the countries between 0 to 1 based on their performance in the key areas of health, education and access to resources.
- It is the sum total of the weights assigned to health, education and access to resources. Indicator to access health is life expectancy at birth, to access knowledge it is adult literacy rate and gross enrollment ratio and resources are measured in terms of purchasing power

Human Poverty Index

This index measures the shortfall in human development. The index is based on the probability of not surviving till 40, adult literacy rate, people not having access to clean drinking water and number of underweight children.

United Nations Development Programme

The United Nations Development Programme (UNDP) since 1990, publishes the Human Development Report by measuring the Human Development Index and Human Poverty Index.

Gross National Happiness

This is Gross National Happiness (GNH) which is another measure of accessing human development and Bhutan is the only country in the world to measure the country's progress through GNH. The GNH encourages the qualitative aspect of development.

International Comparisons

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- On comparing the human development of different countries, it is revealed that the size of the territory and per capita income are not directly related to human development.
- For comparisons, the countries are classified into four groups on the basis of human development scores earned by them.

Very High Level of Human Development

- The countries scoring above 0.802 are categorised under it. There are 49 countries in this group.
- Top ten countries with very high value index are Norway, Australia, Netherlands, United States, New Zealand, Canada, Ireland, Liechtenstein, Germany and Sweden.
- The countries in this group provide education and healthcare which is an important government priority as well as invest a lot in the social sector.

High Level of Human Development

- Countries scoring between 0.702 and 0.801 are grouped under it. They are 56 in number.
- Good governance and lot of investment in social sector has led to high level of human development in these countries.

Medium Level of Human Development

- Countries scoring between 0.555 and 0.701 are grouped under it and there are 39 countries in this group. ‘
- Many of these countries were former colonies and are facing political instability as well as high social diversity.

Low Level of Human Development

- Countries scoring below 0.554 are grouped under it and there are 44 countries in it. These countries are going through political turmoil, social instability, civil war, famine or high incidence of diseases.
- Internal comparisons reveal that the culture, religion or community are non-determinants of human development. Rather, pattern of government expenditure on the social sector, political environment, amount of freedom people have and the distribution of resources play a decisive role in the level of human development.

Chapter 5 - Primary Activities

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Hunting and Gathering

- Hunting and gathering are the oldest primitive activities of man. Early man lived in forests and sustained his life by hunting animals. He has also been gathering various products from the forests to satisfy his needs in early days. Thus, early men sustained on edible plants which they hunted and edible plants which they gathered from the forests.
- Early human beings practised these activities with the help of primitive tools, made of stones, twigs or arrows, so the number of animals killed was limited. But now modern equipments led to the widespread poaching of animals due to which hunting is declared illegal.
- Gathering is practised in high latitude zones of North Canada, Northern Eurasia and Southern Chile, low latitude zones of Amazon Basin, tropical Africa, interiors of South-East Asia. This activity requires small capital investment and low level of technology. Valuable plants, leaves, bark, etc are collected and after simple processing they are sold in local markets. These products cannot compete the world market as synthetic products are lower in price and of better quality.

Pastoralism

- This means domestication and rearing of animals on pastures. This is carried either at subsistence level known as nomadic herding or at commercial level known as commercial livestock rearing.

Nomadic Herding

- Also called pastoral nomadism, it is a primitive activity where herders move with their livestock from one place to another in search of food and water. They rely on animals for food, clothing, shelter, tools and transport.
- Depending on geographical factors different animals are reared in different regions like cattle in tropical Africa, sheep, goat and camel in Sahara, Asiatic deserts, Yak, llamas in Tibet and Andes, reindeer in Arctic and Sub-Arctic areas. Other areas include Mongolia, Central China, Eurasia, South-West Africa and Madagascar.

Transhumance

This is the process of migration from plain areas to mountains during summers and from mountains to plain areas during winters. The tribes of Bakarwals, Gaddis, Bhotiyas, Gujjars of Himalayas and nomadic herders of Tundra region do transhumance.

Commercial Livestock Rearing

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- This practise is associated with western culture, it is more organised and capital intensive. This activity is done on vast ranches that are permanent.
- Modern scientific technology with emphasis on breeding, genetic improvement, disease control and health care is done. Production of meat, wool, hides is done and exported to different world markets.
- Sheep, cattle, goat and horses are reared in the ranches. Australia, New- Zealand, Argentina, Uruguay, South-West Africa, Western United States and Central Asia are major centers of commercial livestock rearing.

Agriculture

Agriculture is practised under multiple combinations of physical and socio- economic conditions, which gives rise to different types of agriculture systems.

Based on methods of farming, different type of crops are grown and livestock raised. The following are the main agricultural systems:

Subsistence Agriculture

In this type of agriculture, farmers consume all or nearly all of the produce as the size of the farm is very small. It can be divided into primitive subsistence agriculture and intensive subsistence agriculture.

Primitive Subsistence Agriculture This is widely practised in tropical areas of Africa/South, Central America and South-East Asia. Different names are given to it in different places like Milpa, Jhuming, Ladang. It is also called slash and burn agriculture or shifting agriculture. Cultivation is done through primitive tools by clearing a patch of land. After 3 to 5 years, when the soil loses its fertility then new land is cleared for cultivation.

Intensive Subsistence Agriculture This type of agriculture is practised in densely populated regions of monsoon Asia. This type of cultivation is either dominated by wet paddy cultivation and crops other than paddy. The per unit area is high but productivity is low as land holdings are small due to high density of population, less use of machines, family labour involved and farm yard manures are used. Crops other than paddy are wheat, soybean, barley, sorghum grown in Northern China, Manchuria, North Korea and Japan. In India, it is practised in Indo-Gangetic plains, dry parts of Western and Southern India.

Plantation Agriculture

This type of agriculture was introduced by the Europeans like cocoa and coffee in West Africa by French, Tea in India and Sri Lanka, rubber in Malaysia, sugarcane and banana in West Indies by British, coconut, sugarcane in Philippines by Spanish, etc.

Here, agriculture is done on large estates or plantations, requires huge capital investment and scientific methods of cultivation.

Extensive Commercial Grain Cultivation

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- This is practised in the interiors of semi-arid parts of mid latitudes like Eurasian steppes, Canadian and American Prairies, Pampas of Argentina, velds of South Africa, Australian Downs and Canterbury plains of New Zealand.
- Crops like wheat, corn, barley, oats and rye are grown. Agriculture is mechanised as farms are very large. There is low yield per acre but high yield per person.

Mixed Farming

- This is practised in North-Western Europe, Eastern-North America, parts of Eurasia and temperate latitudes of Southern continents. Here crop cultivation along with animal husbandry is practised. Crops like wheat, barley, oats, rye, maize, fodder and root crops along with the rearing of farm animals like cattle, sheep, pig, poultry is done to provide extra income.
- There is crop rotation and inter cropping along with high capital expenditure on farm inputs.

Dairy Farming

- It is practised near urban and industrial centers of North-Western Europe, Canada, South-Eastern Australia, New Zealand and Tasmania. It is highly capital intensive due to cost of feeding, milching machines, animal sheds, storage facilities, veterinary services, etc.
- It is also labour intensive as rigorous care of animals is also needed. There is no off season during the year as in case of crop raising.

Mediterranean Agriculture

It is practised on either side of Mediterranean in Europe, North Africa from Tunisia to Atlantic coast, Southern-California, some parts of Chile, South-Western parts of South Africa and South-Western parts of Australia. It is highly specialised commercial agriculture and important supplier of citrus fruits.

Viticulture

This refers to cultivation of grapes in the Mediterranean regions. Inferior grapes are made into raisins and currants while superior grapes are processed into wines. Figs and olives are also cultivated.

Market Gardening and Horticulture

- It is practised in highly developed and industrial regions of North-West Europe, North-East United States and Mediterranean regions. Netherlands specialises in cultivation of Tulip flowers.
- This type of cultivation refers to high value crops like vegetables, fruits and flowers solely for urban markets. Farms are small and mechanised. It is both capital and labour intensive.

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Truck Farming

This refers to the specialisation in the cultivation of vegetables only. The distance between the farms and urban markets can be covered through trucks overnight, hence its name is known as 'truck farming.'

Factory Farming

This refers to livestock rearing mainly poultry and cattle. This is capital intensive, requires maintenance, supervision and scientific breeding. It is practised in industrial regions of Europe and North America.

Co-operative Farming

Under this farming group of farmers form a co-operative society by pooling their resources voluntarily for more efficient and profitable farming. There is help from societies in buying and selling farm products. Individual farms remain intact. It is practised in Denmark, Netherlands, Belgium, Sweden and Italy.

Collected Farming

- It is based on social ownership of the means of production and collective labour. Here farmers pool their resources, though they are allowed to keep a small part with themselves.
- Produce is sold to state at prefixed prices. Members paid according to the nature of work done. It was practised first in Kelkhos in erstwhile Soviet Union.

Mining

It refers to the extraction of mineral resources from the Earth. Its development began with the industrial revolution.

Factors affecting Mining Activity

There are two factors that affect the mining activity. These are:

1. Physical factors related to size, grade and mode of occurrence of the deposits.
2. Economic factors related to demand for mineral, technology, capital to develop infrastructure, labour and transport cost.

Methods of Mining

It is of two types i.e. surface and underground mining. Surface of open cast mining occurs close to the surface, it is easiest and cheapest. The other is underground mining in which minerals are extracted from very deep mines using vertical shafts. It requires lifts, drills and very efficient technology.

The developed economies are retreating from mining due to high labour cost while developing economies are becoming major exporters of minerals due to their large labour force and striving for better standard of living.

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Chapter 6 - Secondary Activities

Manufacturing

The literally meaning of manufacturing is to make by hand, but in the present context, the manufacturing means the conversion of raw material into more useful and valuable fabricated articles with the help of machines.

Manufacturing Industries

These are geographically located manufacturing units that transform raw materials into finished goods of higher value for sale in local or distant markets. The term industry is comprehensive and can also be used in many secondary activities which are not carried on in factories like entertainment industry, tourism industry, etc.

The manufacturing industries are characterised by the following:

- **Specialisation of Skills** In industries, one task is done repeatedly that gives specialisation of doing that task. This involves high cost of manufacturing. On the other hand, mass production involves production of large quantity of standardised parts by each worker performing only one task repeatedly.
- **Mechanisation** The industries use automated processes or machines which does the major production. Human thinking is not required in mechanisation.
- **Technological Innovation** Latest technology is used and constant innovation is done to eliminate waste, quality control, combat pollution and bring efficiency.
- **Organisational Structure and Stratification** Modern manufacturing is characterised by complex machine technology, extreme specialisation, division of labour, vast capital, large organisations and executive bureaucracy.
- **Uneven Geographic Distribution** The industries are concentrated in regions that are rich in mineral and other resources. These areas cover less than 10% of the world's land area. These regions have become the major centres of economic and political power.

Location of industry

The location of industry at a particular place is governed by a large number of geographical and non-geographical factors. Industries maximise profits by reducing costs. Thus, industries should be located at points where the production costs are minimum. Following factors influence the location of industry at particular places:

- **Access to Market** Areas that provide large markets for finished industrial goods like developed areas of Europe, America, Japan, Australia, South Asia have huge concentration of industries.

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- **Access to Sources of Energy** Coal, petroleum and hydroelectricity are main sources of energy. Industries using more power are located close to these sources.
- **Access to Raw Material** Industries based on cheap, bulky and weight-losing materials (ores) like steel, sugar are based close to sources of raw materials. Similarly, processing of dairy products, perishable foods and agro based are done near the sources of raw materials.
- **Access to Labour Supply** Industries are located where there is availability of skilled labour. Some types of manufacturing still require skilled labour.
- **Access to Source of Energy** Industries which use more power are located nearer to the source of energy supply such as iron and steel industries. Energy is most essential to run machines in industries. The main power resources are coal, petroleum, hydroelectricity, natural gas and nuclear energy.
- **Access to Transportation and Communication Facilities** Industries are located in places that have efficient transportation facilities and communication services for the exchange and management of information.
- **Access to Agglomeration Economies** Agglomeration economies refer to the benefits derived from the linkages that exist between different industries. The small industries or ancillary units like to operate near leader industries to benefit from nearness to big or basic industries.
- **Government Policy** For the balanced economic development, governments promote various regions by setting up industries in a particular link between industrial areas.

Classification of Manufacturing Industries

Industries are classified on the basis of their size, inputs/ raw materials, output/ products and ownership.

Industries Based on Size

Based on amount of capital invested, a number of workers employed and volume of production, industries are classified into the following:

Household Industries or Cottage Manufacturing It is the smallest manufacturing unit. Artisans use local raw materials, simple tools and production is done with the help of family members.

Production is done for local consumption and local markets. There is not much capital needed, e.g. mats, baskets, pottery, jewellery, artefacts and crafts.

Small Scale Manufacturing These type of industries employ semi-skilled labour, operate by power driven machines, use local raw materials and manufacture products in workshops. It provides employment and raises local purchasing power. India, China, Indonesia and Brazil have developed labour intensive small scale manufacturing units.

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Large Scale Manufacturing Here mass production takes place, involves large market, many raw materials, huge energy requirements, specialised workers, advanced technology and large capital. Large scale manufacturing industries are divided into two parts i.e. Traditional large scale industrial regions and High technology large scale industrial regions.

Industries Based on Inputs/Raw Materials

On the basis of raw materials used, industries are classified as follow:

- **Agro based Industry** This involves processing of raw materials from the fields and farms into finished products like sugar, fruit juices, beverages, oils and textiles (cotton, jute, silk), rubber, etc.
- **Food Processing** This is part of agro based industry and includes processes like canning, producing cream, fruit processing, confectionery, drying, fermenting and pickling.
- **Agri Business** This is commercial farming on an industrial scale. The farms are mechanised, very large and highly structured, like tea plantation and tea factories near the plantations.
- **Mineral based Industry** These are industries that use minerals as raw materials such as ferrous like iron and steel and non-ferrous like aluminium, copper, etc. Mineral based non-metallic industries are of cement and pottery.
- **Chemical based Industry** These industries use natural chemical minerals like salts, sulphur, potash, mineral oil in petrochemical industry and chemicals obtained from wood and coal. Synthetic fiber and plastics are other examples of chemical based industry.
- **Forest based Industry** Industries that use forest products such as timber, wood, bamboo, grass, lac, etc come under forest based industry.
- **Animal based Industry** Industries that use animal products such as leather, woollen textiles, ivory are grouped under animal based industry.

Industries based on Output/ Product

This refers to industries based on the finished products or output. These are:

1. **Basic Industries** These are the industries that produce raw material to be used in other industries such as iron and steel.
2. **Consumer Goods Industries** These are the industries which produces what is consumed by consumers directly such as tea, biscuits, toiletries, etc.

Industries based on Ownership

Based on the ownership, the industries are grouped as:

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- **Public Sector Industry** This refers to industries that are owned and managed by government. In India, it is called public sector undertakings. Socialist economies have all state owned industries.
- **Private Sector Industry** This refers to industries that are owned by private individuals and also managed by them. Capitalist economies have mostly private owned industries.
- **Joint Sector Industry** Industries that are jointly owned and managed by joint stock companies or established by private and government sector are called joint sector industries.

Foot Loose Industries

These industries do not depend on any specific raw material so they can be located at any place. They largely depend on component parts, employ small labour force and produce in small quantity.

Traditional Large Scale Industrial Regions

Traditional large scale industries are mostly heavy industries located near coal fields and involved in metal smelting, heavy engineering, chemical manufacturing or textile production. Their features are high employment, high density of housing but poor services, unattractive environment, pollution and waste heaps. Due to these problems, many industries are closed leading to unemployment, emigration and wastelands.

Smokestack Industry

This is the other name given to traditional large scale industries as they emit a lot of smoke and pollute the environment.

The Ruhr Coal Field, Germany

- This area was a major industrial region due to coal and iron-ore deposits. But the industry started shrinking as demand of coal declined, iron-ore exhausted, industrial waste and pollution increased.
- Now a **New Ruhr** landscape has emerged that focusses on other products like Opel car assembly plant, new chemical plants, universities and out of town shopping centres.

High Technology Industry

- Also called high-tech industry, it is highly technical and incorporates advanced scientific and engineering research and development strategy.
- The workforce are highly skilled specialists, professionals (known as white collar) who outnumber the production labour (blue collar).
- Robotics, computer aided design and manufacturing, electronics, new chemicals and pharmaceuticals are examples of these industries.

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Technopolis

This refers to high-tech industries that are concentrated regionally, self sustained and highly specialised, e.g. Silicon valley near San Francisco and Silicon forest near Seattle.

Iron and Steel Industry

- It is known as basic industry as it provides raw materials or base to other industries. It is also called heavy industry due to its bulky raw material and heavy finished products.
- These industries are located near the source of raw materials i.e. iron ore, coal, manganese and limestone or near ports where it could be early brought.

Mini Mills

As compared to large integrated steel plants, mini mills are gaining importance as they make steel from scrap metal that is present in abundance.

Distribution

This industry is spread in developed and developing countries such as America, UK, Germany, France, Belgium, Ukraine, Japan, China and India (Jamshedpur, Durgapur, Raurkela, etc.)

Cotton Textile Industry

This industry has three sub-sectors:

1. **Handloom** This is labour intensive, employs semi-skilled workers, requires small capital and involves processes like spinning, weaving and finishing of the fabrics.
2. **Powerloom** This is less labour intensive, uses of machines and production is more.
3. **Mill Sector** This is highly capital intensive, requires good quality raw cotton and produces in bulk.
India, China, USA, Pakistan, Uzbekistan and Egypt produce more than half of world's raw cotton. Now the cotton textile industry is shifting to less developed countries due to labour cost.

Chapter 7 - Tertiary and Quaternary Activities

Types of Tertiary Activities

There are four types of tertiary activities. They are trade, transport, communication and services. These include provision of services in exchange of payments.

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Trade and commerce

Trade and commerce is essentially buying and selling of items produced elsewhere. The collection and distribution points where trading takes place are called trading centres. These centres are divided into:

- **Rural Marketing Centres** They are quasi urban and cater to local needs and areas. Most of these have mandis (wholesale markets) and retail markets. In rural areas, there are **periodic markets** that may be weekly or bi-weekly and people from the nearby areas meet their demands. These markets are held on specified dates and shopkeepers move from one place to another.
- **Urban Marketing Centres** These markets sell ordinary as well as specialised goods and services, e.g. markets for labour, housing, semi or finished products. Services of educational institutions and professionals like teachers, doctors, lawyers also develop.
- **Retail Trading** In this type of trading, goods are directly sold to consumers. This trading is done through fixed establishments or stores, small shops, consumer cooperatives, big departmental stores and chain stores. The chain stores buy commodities in bulk and then hire skilled specialists for executive tasks. Street peddling, handcarts, trucks, door-to-door, mail order, telephone and Internet are examples of non-store retail trading.
- **Wholesale Trading** Here bulk buying takes place directly from the manufacturer by numerous intermediary merchants. The merchants/ wholesalers extend credit to retailers.

Transport

Transport is a tertiary activity in which people, materials and manufactured goods are physically carried from one place to another. While selecting the mode of transport, distance, time and cost are seen.. Distance can be measured as km distance, time taken to travel particular route as time distance, expense of travelling on a route as cost distance.

Factors Affecting Transport

Demand and routes are two major factors which affect transport services.

- **Demand** Transport depends on the size of population and standard of living of people. The larger the population size, the greater is the demand for transport.
- **Route** It refers to the transport network depend on location of cities, towns, villages, industrial centres, availability of raw materials, nature of landscape, type of climate, availability of funds, etc.

Network

A network is a well developed transport system that is made up of nodes and links. A node is a meeting point of two or more routes and every road that joins any two nodes is called a link.

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Communication

Communication services involve in the transmission of words, messages, facts and ideas. The development of transport facilitated communication as messages were carried by hand, animals, boat, road, rail and air. But new technology has made communication independent of transport, such as mobiles, telephony and satellites. Some of the communication services are discussed below:

- **Telecommunications** The development of modern technology has revolutionised communication and it has become direct and instantaneous, e.g. telegraph, morse code and telex in last century and now satellites, mobile, telephony, etc are used.
- **Mass Media** Communication means through which messages could be sent to vast audiences around the world are called mass media, e.g. radio, television, newspapers, etc. The Internet has revolutionised the global communication.

Services

There are different levels at which services are provided and availed. Some are meant for industry, some for people, and some for both industry and people i.e., transport. Services can be divided into three sub-categories. They are:

- **Low Order Services** It includes common and widespread services like grocery shops, laundries, etc.
- **Domestic Services** It includes housekeepers, cooks and gardeners which migrate from rural areas in search of employment.
- **High Order Services** These are specialised and less common like accountants, consultants and physicians. Some services are supervised and/or regulated by government like making, maintaining highways, bridges, fire fighting departments, education, healthcare, etc.

Thus, services are present in organised sector that is government owned or big corporations. Some are present in unorganised sector like low order and domestic services. Mumbai's dabbawala in India is one such service of unorganised sector.

People Engaged in Tertiary Activities

Earlier more number of people were employed in the primary and secondary sector as these sectors provided more jobs. But, now there has been a shift of jobs to tertiary or service sector. In developed countries, a higher percentage of workers are employed in providing services as compared to less developed countries.

Some Selected Examples

Some of the selected examples that are related to the people engaged in tertiary activities are as follows: %

Tourism

Tourism is part of service sector that refers to travel undertaken for purpose of recreation rather

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than business. This industry generates jobs as people are engaged in providing accommodation, meals, transport, entertainment, infrastructure retail trading and crafts.

Tourist Regions

- Tourism can be seasonal or throughout the year like warmer places around the Mediterranean coast, West coast of India during winters, mountains in summers or winter spots regions found mainly in mountainous areas.
- Historic towns, religious places, heritage sites offer tourism throughout the year. Factors Affecting Tourism
- The rise in tourism industry is due to increased demand for it which is thus influenced by improvement in standard of living and increased leisure time.
- Another factor is improvement in transportation that has made travel easier and destinations reachable.

Tourist Attractions

Tourist attractions are specific features of a place that attract people. These are as follows:

- **Climate** In winter holidays, areas having warm sunny weather is preferred like beaches in Southern Europe, so it attracts more number of tourists there.
- **Landscape** Mountains, lakes, spectacular sea coasts and landscapes not completely altered by man are good tourist attractions.
- **History and Art** Ancient or picturesque towns, archaeological sites, historically important places having castles and palaces attract tourists.
- **Culture and Economy** Areas having rich cultures attract people as they go their to experience ethnic and local customs. Places giving economic benefits are also attractions such as cheap home stays in Goa, Madikere and Coorg in Karnataka.

Medical Services for Overseas Patients in India

Medical services or tourism takes place when medical treatment is combined with international tourism activity. People from developed countries like US are visiting India for medical tourism or services. This brings economic benefits to India and other countries where medical tourism is taking place like Thailand, Singapore and Malaysia.

Other medical related activities such as outsourcing of medical tests, data interpretation, reading radiology images, interpreting Magnetic Resonance Images (MRIs) and ultrasound tests are taking place in India, Australia and Switzerland.

Quaternary Activities

The activities related to knowledge oriented, involving collection, production and dissemination of

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information come under quaternary activities. They centre around research development and may involve specialised knowledge and technical skills.

Software developers, mutual fund managers, doctors, accounting, brokerage firms are some examples of quaternary activities. They can be outsourced even as these are not tied to resources or affected by the environment or markets. This sector has replaced primary and secondary sector and absorbs half of the population in developed economies.

Quinary Activities

The activities that are highly specific and specialised are placed under it. These include high level decision-makers, administrators, government officials, research scientists, etc. They are also known as gold collar professions.

Outsourcing

- It means contracting out or giving work to an outside agency to improve efficiency and to reduce cost. It is termed as off-shoring when work is transferred to overseas location.
- Outsourcing provides employment in developing Countries of India, China, Botswana, etc. Information technology, human resources, customer support, call centres, data processing and other IT related services are examples of outsourcing.

BPO

- BPO stands for Business Process Outsourcing which involves customer support services, call centres and similar other processes.
- The developed countries transfer these jobs as overhead costs are much lower making it profitable to get job work carried out.

KPO

KPO stands for a Knowledge Process Outsourcing which involves skilled workers and enables companies to create additional business opportunities, e.g. e-learning, business research, etc.

The Digital Divide

- Development emerging from the information and communication technology is unevenly spread across the globe. Some regions have prospered while others are lagged behind. This is known as digital divide.
- In developing countries, this kind of divide is seen more than developed countries. Here, the metropolitan cities are developed more than rural areas.

Chapter 8 - Transport and Communication

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Transport

Transport is a facility or a service for the carriage of persons and goods from one place to another using humans, animals and different kinds of vehicles. It is an organised service industry that handles loading, unloading and delivery.

Modes of Transportation

- The main modes of transportation are land, water, air and pipelines. These are used for inter-regional and intra-regional transport, and each one (except pipelines) carries both passengers and goods.
- Several places (nodes) join together by a series of routes (links) to form a pattern is called transport network.

Land Transport

This includes movement of goods and services over land i.e. roads and rails. Earlier humans themselves were carriers such as in palanquin (palki or doli), later pack animals such as mules, horses, camels were used. Dogs and reindeers were used in North America. In India, bullocks were used to pull carts. The revolution came after invention of steam engine in 18th century that started railways and roadways with the invention of internal combustion engines.

Among the new means of land transport are pipelines, ropeways and cableways. Rope and cableways are generally found on steep mountain slopes and mines, which are not suitable for building roads.

Roads

It is most economical for short distance and gaining importance for freight transport due to its door to door service. Metalled roads are all weather roads while unmetalled roads are not serviceable in all seasons due to their simple construction. Though heavy rains and floods make both the roads unserviceable. The quality, construction and maintenance of roads is better in developed countries than in developing countries as it requires heavy expenditure.

Road Density

- The total motorable road length of the world is only about 15 million km, in which North America separately accounted 33%. Although, North America in compare to Western Europe registered highest number of vehicles as well as highest road density.
- Road density is the total length of roads per hundred square kilometre of area.

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Countries and their Road Density

Countries	Density (For every 100 km ² area)
Japan	327
UK	162
France	164
Sri Lanka	151
India	105
Spain	68
USA	67

Traffic Flows

It refers to traffic on roads that has increased dramatically in recent years. As the road network cannot cope with the demands of traffic, so congestion occurs.

Highways

- They are metalled roads connecting distant places for unobstructed vehicular movement. These are 80m wide with separate traffic lanes, bridges, flyovers and dual carriageways
- In North America, there is dense network of highways. Pacific coast is linked to Atlantic coast, Vancouver is connected to Newfoundland by Trans-Canadian highway and Edmonton is connected to Anchorage through Alaskan highway.
- Trans-continental Stuart highway connects Darwin, Melbourne to Alice springs in Australia. Europe has well developed highway network. Moscow-Vladivostok highway is important for Russia. Highways criss-cross the country in China.
- In India, National Highway No.7 (NH7) connecting Varanasi and Kanyakumari is the longest highway of the country. (The golden quadrilateral or super expressway is under construction). Now, NH-44 became the longest running highway in India. It connects Srinagar to Kanyakumari. NH-7 is renamed as National Highway 44. In Africa, Algiers in North is connected to Guinea and Cairo connected to Capetown in South.

Border Roads

These are roads laid along international boundaries. These roads help in transport of goods to border areas and military camps.

Railways

Railways are best suited for the transportation of bulky goods and passengers over long distances. Highly industrialised areas, urbanised regions and mineral rich areas are linked to railways for the transportation of ores, grains, timber and machinery. All the continents have dense network while Europe is having the highest density of railways. The railway network of Africa, Asia and South

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America connects the mineral rich and fertile areas and is developed primarily to utilise the natural resources.

Trans-Continental Railways

The railway line that runs across the continent and links its two ends is called a trans-continental railway line. They are constructed for economic and political reasons.

The following are the most important of these:

Trans-Siberian Railway

It is in Russia and the longest railway in the world. It runs between St. Petersburg in West to Vladivostok in East, passing through Moscow, Irkutsk, Chita, etc. It links important agro centre with fur centre connecting rail routes to important cities of Asia.

Trans-Canadian Railway

Constructed in 1886, it is 7050 km long railway in Canada that links Halifax in East to Vancouver in West. It passes through the industrial region connecting the wheat belt of Prairies and the coniferous forest area so it is considered economic artery of Canada. Wheat and meat are the important exports on this route.

The Union and Pacific Railway

This rail line connects New York on the Atlantic coast to San Francisco on the Pacific coast passing through Cleveland, Chicago, Omaha, Evans, Ogden and Sacramento. The most valuable exports on this route are ores, grain, paper, chemicals and machinery.

The Australian Trans-Continental Railway

This rail line runs West-East across the southern part of the continent from Perth on the West coast to Sydney on the East coast passing through Kalgoorlie broken hill and port Augusta. Another North-South line connects Adelaide and Alice spring and to be joined later to the Darwin-Birdum link.

The Orient Express

This line runs from Paris to Istanbul passing through Strasbourg, Munich, Vienna, Budapest and Belgrade. It has reduced the 10-day journey to only 4 days. Cheese, wine, bacon, oats, fruits and machinery are chief exports on this rail route.

Water Transport

This is the cheapest mode of transport as no construction cost is there and very little maintenance cost. The linking of oceans have brought greater navigation with ships of various sizes. Water transport is divided into sea routes and inland waterways.

1. Sea Routes

Sea and oceans provide smooth highway traversable in all directions with no maintenance costs. Modern passenger ships and cargo ships are equipped with various navigation aids. The important sea routes are as follows:

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- **North Atlantic Sea Route** It links North-Eastern USA and North-Western Europe. It is the busiest in the world and also called Big Trunk route.
- **Mediterranean-Indian Ocean Sea Route** This route connects industrialised Western Europe with West Africa, South Africa, South-East Asia, Australia and New Zealand. Natural resources such as gold, diamond, copper, tin, groundnut, oil palm, coffee and fruits are transported through it.
- **Cape of Good Hope Sea Route** This route links West Europe and West African countries with Brazil, Argentina and Uruguay in South America. Traffic is less on this route because the countries falling in this route have similar products and resources.
- **The North Pacific Sea Route** This route connects the ports on the West coast of North America with those of Asia. These are Vancouver, Seattle, Portland, San Francisco and Los Angeles of American side with Yokohama, Kobe, Shanghai, Hong Kong, Manila and Singapore of Asian side.
- **The South Pacific Sea Route** This route is used for reaching Hong Kong, Philippines and Indonesia and also links Western Europe and North America with Australia, New Zealand and Pacific Islands via the Panama Canal. Panama is 12000 km far from Sydney. Honolulu is an important port on this route.
- **Coastal Shipping** Coastal shipping is a convenient mode of transportation with long coastlines, e.g. USA, China and India. This type of shipping can reduce congestion on land routes.

Shipping Canals

There are two canals that serve as gateways of commerce for both the Eastern and Western worlds. They are:

- **The Suez Canal** Constructed in 1869, it is a man-made canal linking the Mediterranean sea and the Red sea. It is 160 km long and 11 -15 m deep without any locks and sea water flows freely through it.
- **The Panama Canal** It is a man-made canal linking Atlantic ocean with Pacific ocean. It is 72 km long and involves a deep cutting for a length of 12 km and has 6 locks. It shortens the distance between New York and San Francisco by 13000 km by sea.

2. Inland Waterways

Rivers, canals, lakes and coastal areas are inland waterways for the transportation of cargo and passengers. The development of inland waterways depends on navigability, water flow and transport technology. The important inland waterways are: The

- **Rhine Waterways** This waterway links the industrial areas of Switzerland, Germany, France, Belgium and the Netherlands with the North Atlantic sea routes. The river Rhine flows

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through Germany and Netherlands. It flows through a rich coal field, Dusseldorf is an important port in this region. This waterway is the world's most heavily used. More than 20,000 ocean-going ships and 200,0 inland vessels move from this waterways every year.

- **The Danube Waterway** The Danube river which is navigable upto Tauma Severin, emerges in the Black Forest. It is used mainly for the export of wheat, maize, timber and machinery.
- **The Volga Waterway** Volga is one of the developed waterways of Russia. It is navigable upto 11,200 km and drains into the Caspian Sea. It is connected to Moscow region and the Black Sea through Volga-Moscow canal and Volga-Don canal respectively.
- **The Great Lakes-St Lawrence Seaway** The Great Lakes alongwith estuary of St Lawrence river form a waterway in North America. Duluth and Buffalo are two important ports on this route.
- **The Mississippi Waterways** The Mississippi-Ohio waterway links the interior part of USA with the Gulf of Mexico in the South. Large steamers can move upto Minneapolis.

Air Transport

It is the fastest means of transport but it is very costly. Air transport has brought connectivity revolution in inhospitable deserts, mountainous regions and snow fields where other means of transport are not reachable. Due to high construction and maintenance cost, air transport is more developed in highly industrialised countries. Supersonic aircrafts cover the distance in very short time.

Inter- Continental Air Routes

USA accounts for 60% of airways of the world. Important cities are nodal points where air routes converge or radiate to all continents. Africa, Asiatic part of Russia and South America lack air services, sparser population or limited landmass or low economic development.

Pipelines

These are used to transport water, petroleum, natural gas, liquidified coal for an uninterrupted flow. Milk is supplied through pipelines in New Zealand. USA has dense network of pipelines. Big Inch is famous pipeline of USA that transports petroleum from the oil wells of the Gulf of Mexico to the North-Eastern states. In Europe, Russia, West Asia and India, oil wells are linked to refineries through pipelines.

Communications

Long distance communication in the form of telegraph and telephone are important. In mid-19th century, American Telegraph and Telephone company (AT&T) enjoyed monopoly. In developing countries the use of cell phones has gained importance for rural connectivity. Latest technology developments have resulted in Optical Fibre Cables (OFC). They allow large

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quantities of data to be transmitted that are virtually error free. Now the telecommunication merged with computers to form integrated networks termed as Internet.

Satellite Communication in India

Artificial satellites are deployed in Earth's orbit to enhance communication and improve connectivity. This is satellite communication which has reduced the per unit cost and time of communication also.

India developed its own satellite Aryabhata and launched it on 19th April, 1979, Bhaskar -I in 1979 and Rohini in 1980. Bhaskar, Challenger and INSAT-IB satellites are used for long distance communication and weather forecasting.

Cyber Space-Internet

This is the latest technology in which there is instant connectivity by accessing the electronic computerised space. It is called cyber space or Internet and is encompassed by the World Wide Web (www).

Majority of Internet users are in USA, UK, Germany, Japan, China and India. The social and economic space has expanded through e-mail, e-commerce, e-learning and e-governance.

Chapter 9 - International Trade

History of International Trade

- In ancient times, trade was restricted to local markets. Slowly long distance trade developed; of which Silk Route is an example. The route was 6000 km long connecting Rome to China and traders transported Chinese silk, Roman wool, metals, etc through this route. Later, sea and ocean routes were discovered and trade grew.
- The Slave Trade emerged in 15th century in which the Portuguese, Dutch, Spaniards and British captured African natives and sold to plantation owners in America. After Industrial Revolution, industrialised nations imported raw materials and exported finished products to non-industrial nations.
- International trade is the result of specialisation in production and division of labour. It is based on the principle of comparative advantage that is mutually beneficial to trading partners.

Basis of International Trade

The factors on which international trade depends are as follows:

- **Difference in National Resources** The resources are unevenly distributed in the world. These differences mainly refer to geology, mineral resources and climate.

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- **Geological Structure** This means the relief features, type of land such as fertile, mountainous, lowlands, that support agriculture, tourism and other activities.
- **Mineral Resources** The regions rich in minerals will support industrial development that leads to trade.
- **Climate** It influences the type of flora and fauna that is found in a region, such as wool production in cold regions. Cocoa, rubber, Bananas can grow in tropical regions.
- **Population Factors** The size, distribution and diversity of population between countries affect the trade in respect of type and volume of goods. Large volume of internal trade than external trade takes place in densely populated areas due to consumption in local markets.
- **Cultural Factors** Distinctive forms of art and craft develop in certain cultures and give rise to trade e.g. porcelain and brocades of China, carpets of Iran, Batik cloth of Indonesia, etc.
- **Stage of Economic Development**
Industrialised nations export machinery, finished products and import foodgrains and raw materials. The situation is opposite in agriculturally important countries.
- **Extent of Foreign Investment** Developing countries lack capital so foreign investment can boost trade in developing countries by developing plantation agriculture.
- **Transport** Lack of transport in older time restricted trade only to local areas. The expansion of rail, ocean and air transport, better means of refrigeration and preservation, trade has experienced spatial expansion:***'

Aspects of International Trade

There are three very important aspects of international trade:

- **Volume of Trade** It is measured simply as the total value of goods and services traded. However, actual tonnage of traded goods makes up the volume but services traded cannot be measured in tonnage.
- **Composition of Trade** Earlier primary' goods were more in total traded goods, then there was dominance of manufactured goods and now there is dominance of service sector which includes transportation and other commercial services.
- **Direction of Trade** Earlier valuable goods and artefacts were exported to European countries by the developing countries. Later in the 19th century, manufactured goods from European countries were exchanged with foodstuffs and with raw materials from their colonies.

Types of International Trade

There are two types of international trade:

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- **Bilateral trade** It is between two countries when they enter into an agreement to trade certain goods in which they are specialised.
- **Multilateral trade** It is conducted with many trading countries at the same time at goods which the countries are specialised in. The country may also grant the status of Most Favoured Nation (MNF) on some of trading partners.

Balance of Trade

- It refers to the volume of goods and services imported and exported by one country to other countries. Favourable balance of trade means the value of exports is more than its imports.
- Unfavourable balance of trade means that imports are more than exports. Balance of payments affects a country's economy as negative balance means country's expenses are more than its income.

Case for Free Trade

- Free trade or trade liberalisation is the act of opening up of economics so that more trade takes place. This is done by bringing down trade barriers like tariffs. But trade liberalisation causes competition and can cause dumping.
- Dumping is the selling of a commodity in two countries at a price that differs for reasons not related to costs. Countries need to be cautious about dumped goods.

World Trade Organisation [WTO]

- General Agreement for Tariffs and Trade (GATT) was formed in 1948 to make the world free from tariffs as well as non-tariff barriers.
- On 1st January, 1995, the GATT was transformed into the World Trade Organisation to set-up an institution for the promotion of free and fair trade amongst different countries of the world.
- The WTO sets the rules for the global trading system. The headquarters of WTO is located in Geneva, Switzerland and 164 countries are its members.
- However, WTO has been criticised and opposed by those who are worried about the effects of free trade and economic globalisation. They argued that free trade is not beneficial to the ordinary people as it is widening the gap between rich and poor.
- They also argued that issues of health, workers' rights, child labour and environment are ignored.

Regional Trade Blocs

These are developed as a response to the failure of global organisations. There are 120 regional

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trade blocs that generate 52% of the world's trade.

Some of the trade blocs are as follows:

Regional Blocs	Headquarters	Member Nations	Origin	Commodities	Other Areas of Cooperation
ASEAN (Association of South-East Asian Nations)	Jakarta, Indonesia	Brunei, Indonesia, Malaysia, Singapore, Thailand, Vietnam, Cambodia, Laos, Myanmar, Philippines.	August 1967	Agro products - rubber, palm oil, rice, copra Minerals -copper, coal, nickel, tungsten Energy -petroleum, natural gas, software products	Accelerate economic growth, cultural development, peace and regional stability
CIS (Commonwealth of Independent States)	Minsk, Belarus	Armenia, Azerbaijan, Belarus, Georgia, Kazakhstan, Kyrgyzstan, Moldova, Russia, Tajikistan, Turkmenistan, Ukraine and Uzbekistan	-	Crude oil, natural gas, gold, cotton, fibre, aluminium	Integration and cooperation on matters of economics, defence and foreign policy
EU (European Union)	Brussels, Belgium	Austria, Belgium, Denmark, France, Finland, Ireland, Italy, the Netherlands, Luxemburg, Portugal, Spain, Sweden and UK	EEC- March, 1957 EU- February, 1992	Agro products, minerals, chemicals, wood, paper, transport vehicles, optical instruments, clock-work of arts, antiques	Single market with single currency
LAIA (Latin American Integration Association)	Montevideo, Uruguay	Argentina, Bolivia, Brazil, Chile, Cuba, Columbia, Ecuador, Mexico, Paraguay, Panama, Peru, Uruguay and Venezuela	1960	-	-
N A F T A (North American Free Trade Association)		U S A, Canada and Mexico	1994	Agro products, motor vehicles, automotive parts, computers, textile	-
O P E C (Organisation of Petroleum Exporting Countries)	Vienna, Austria	Algeria, Indonesia, Iran, Iraq, Kuwait, Libya, Nigeria, Qatar, Saudi Arabia, UAE and Venezuela	1949	Crude petroleum	Coordinate and unify petroleum policies
S A F T A (South Asian Free Trade Agreement)		Bangladesh, Maldives, Bhutan, Nepal, India, Pakistan, Afghanistan and Sri Lanka.	January 2006	-	Reduce tariffs or inter-regional trade

Concerns Related to International Trade

This can be summarised as merits and demerits of international trade.

- **Merits of International Trade** International trade is beneficial if it promotes regional specialisation, higher level of production, better standard of living, worldwide availability of goods and sendees, equalisation of price and wages and diffusion of knowledge and culture.
- **Demerits of International Trade** The demerits are, it leads to dependence on other countries, uneven levels of development, exploitation and commercial rivalry.

Gateways of International Trade

Harbours and ports are the chief gateways of international trade. These ports facilitate the passage of cargos and travellers as well as provide facilities of docking, loading, unloading and storage.

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Types of Ports

Ports are generally, classified according to the types of traffic which handle. Types of ports on the basis of cargo handled are:

- **Industrial Ports** The ports that handle bulk cargo like grain, ores, oil, chemicals are called industrial ports.
- **Commercial Ports** Ports handling packaged products, manufactured goods, passengers are commercial ports.
- **Comprehensive Ports** Ports that handle bulk and general cargo in large volumes are called comprehensive ports. Most of the world's great ports are classified as comprehensive ports.

Types of Ports on the basis of Location

- **Inland Ports** Ports located away from the sea coasts and linked to the sea through a river or a canal are inland ports, e.g. Mannheim on Rhine river.
- **Out Ports** Ports in deep waters built away from the actual ports and serving big ships are called out ports, e.g. Athens and its out port Piraeus in Greece.

Types of Ports on the basis of Specialised Functions

- **Oil Ports** Ports that deal in the processing and shipping of oil are known as oil ports. These are tanker ports like Tripoli in Lebanon and refinery' ports like Abadan on the Gulf of Persia.
- **Ports of Call** Ports which originally developed as calling points on main sea routes where ships used to anchor for refuelling, watering and taking food items are called ports of call, e.g., Honolulu and Aden.
- **Packet Station** Also known as ferry ports, these are exclusively concerned with the transportation of passengers and mail across water bodies covering short distances, e.g., Dover in England and Calais in France.
- **Entrepot Ports** These are collection centres where the goods are brought from different countries for export, e.g., Singapore is an entrepot for Asia.
- **Naval Ports** These ports serve warships and have repair workshops for them, e.g., Kochi, Karwar in India.

Chapter 10 - Human Settlements

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Rural-Urban Dichotomy (Difference)

The human settlements can be differentiated in terms of rural and urban, population size, nature of activities, structure, functions, etc. There is no uniformity in the differentiation of the settlements. But, the basic difference between towns and villages is that in towns the main occupation of the people is related to secondary and tertiary sectors, on the other side, in villages people are engaged in primary activities.

- **Sub-Urbanisation** It is a new trend of people moving away from congested urban areas to cleaner areas outside the city in search of better quality of life.
- **Census of India, 1991** It defines urban settlements as places which have municipal corporation, cantonment board, notified town area committee and have a population of 5000 persons and above and where 75% of male workers are engaged in non-agricultural activities and density of population is at least 400 persons per sq km.

Types and Patterns of Settlements

Settlements can be classified into compact and dispersed by their shape, pattern and types. The major types classified by shape are as follows:

- **Compact or Nucleated Settlements** In these settlements, large number of houses are built very close to each other and they develop along the river valleys and in fertile plains. There are close knit communities and people here share common occupations.
- **Dispersed Settlements** In these settlements, houses are spaced far apart and often interspersed with fields such as a place of worship, a market that binds the settlement together.

Rural Settlements

These settlements are closely and directly related to land. They are dominated by primary activities like agriculture, fishing, animal husbandry, etc. The factors affecting the location of settlements are as follows:

- **Water Supply** Rural settlements are located near water bodies such as rivers, lakes and springs as water is needed for irrigation, fishing, navigation and drinking.
- **Land** Fertile lands suitable for agriculture are places of human settlement like villages in rolling countryside in Europe avoiding swampy areas, low lying river valleys and coastal plains suited for wet rice cultivation in South-East Asia.
- **Upland** Dry points like uplands, terraces, leaves that are not prone to flooding are places of settlements. In tropical countries, people build their houses on stilts near marshy lands to protect themselves from flood, insects and animal pests.

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- **Building Material** Settlements are made in those places where building materials are available like cave dwellings in China, mud bricks houses in African Savanna and igloos with ice blocks in polar regions.
- **Defence** Places that form good defensive site are developed as settlements like defensive hills, islands, etc. In India, forts were built on hills.
- **Planned Settlements** Planned settlements are constructed by government by providing shelter, water and other infrastructure on acquired lands, e.g. canal colonies in Indira Gandhi Canal Command Area in India.

Rural Settlement Patterns

This refers to the way the houses are sited in relation to each other. The surrounding topography terrain influence the shape.

They are classified as below:

1. On the basis of setting, the main types are plain and size of village, plain villages, plateau villages, coastal villages, forest villages, desert villages, etc.
2. On the basis of functions, there may be farming villages, fishermen villages, lumberjack villages, pastoral villages, etc.
3. On the basis of forms/shapes of the Settlement, the villages are developed in geometrical forms and shapes such as:

Linear Pattern The houses are located along the road, railway line, river, canal edge of a valley, along a levee.

Rectangular Pattern The settlements are located in plain areas or in wide inter montane valleys. The roads are rectangular and cut each other at right angles.

Circular Pattern The settlements develop around lakes, tanks and the central part remains open for keeping the animals to protect them from wild animals.

Star-like Pattern These settlements develop where several roads converge and the houses are built along the roads.

T-shaped, Y-shaped, Cross-shaped or Cruciform Settlements The T-shape settlements develop at tri-junctions, Y-shaped settlements emerge where two roads converge on the third and houses are built along these roads, cruciform settlements develop on the cross-roads and houses extend in all the four directions.

Double Village These settlements extend on both sides of a river where there is a bridge or a ferry.

Problems of Rural Settlements

- Rural settlements in developing countries are large in number and have poor infrastructure. There is inadequate supply of water in these settlements.

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- Water borne diseases like cholera, jaundice, etc are a common problem. There is lack of irrigation facilities, problem of drought and flood in rural settlements. Inadequate sanitation facility, toilet and garbage disposal facilities cause health related problems.
- Proper housing and separate shed for animals are not there. Rural settlements mostly lack metalled roads and modern communication network. Health centres and educational institutions are less in number.

Urban Settlements

There had been a rapid growth of urban settlements around the world. The first city to reach a population of one million was London in AD 1810. At present 48% of the world population live in cities.

Classification of Urban Settlements

Common basis of classification are:

- **Population Size** It refers to the lower limit of the population for a settlement to be designated as urban. It is not universal and varies from country to country. In Columbia, a settlement having population of 1500 is termed as urban, in Argentina and Portugal it is 2000, 2500 in USA and Thailand, 5000 in India, 30,000 in Japan, 250 in Denmark, Sweden and Finland, 300 in Iceland, and 1000 in Canada and Venezuela.
- **Occupational Structure** In some countries, the major economic activities alongwith population size designate a settlement as urban. In Italy, settlement called as urban if more than 50% of its economically productive population is engaged in non-agricultural pursuits. India has set its criterion at 75%.
- **Administration** Administration set up also designates a settlement as urban in some countries. In India, if an area has a municipality, notified area council, then it is considered urban.
- **Location** Location of urban centres is examined with reference to their function, e.g. strategic towns offering natural defence, mining towns, industrial towns, tourist centres, places with historical relics and other places that can provide proper living conditions have the potential to develop into urban centres.

Functions of Urban Centres

On the basis of the functions, the urban settlements are classified into the following:

- **Administrative towns** National capitals having administrative offices like New Delhi, Canberra, London, Beijing, etc are called administrative towns. Provincial (sub-national) towns can also have administrative functions, e.g. Victoria (British Columbia), Albany (New York), etc.

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- **Trading and Commercial Towns** Agricultural market towns like Winnipeg, banking and financial centres like Frankfurt, large inland centres like Manchester, transport nodes like Lahore, Baghdad, Agra are important trading centres.
- **Cultural Towns** Pilgrimage places like Jerusalem, Varanasi, Jagannath Puri, etc are considered cultural towns. Other centres like health and recreation (Miami), industrial (Pittsburgh and Jamshedpur), mining, quarrying (Dhanbad) and transport (Singapore and Mughal Sarai) are also urban settlements.

Classification of Towns on the basis of Forms

Urban settlements can be linear, square, star or crescent shaped. Cities in developed countries are planned while in developing countries have evolved historically with irregular shapes. Chandigarh and Canberra are planned cities while smaller town in India have evolved historically from walled cities to large urban sprawls.

- **Addis Ababa** Established in 1878, it is capital of Ethiopia and is located in hill valley topography. It is a large nodal centre, has large markets and government headquarters. The city has witnessed rapid growth and expansion in all directions.
- **Canberra** Established in 1912, it is the capital of Australia. It is a garden city with wide open spaces, parks and gardens. Initially, it was built to accommodate 25,000 people but now it has expanded to accommodate many satellite towns.

Types of Urban Settlements Problems of Urban Settlements

Depending upon the size and services available, urban centres are classified further as follows:

- **Towns** These can be well understood with reference to 'village'. Specific functions such as manufacturing, retail and wholesale trade, and professional services exist in towns.
- **City** They are larger than towns, have greater number of economic functions, tend to have transport terminals, major financial institution and administrative offices. In the words of Lewis Mumford, "The city is in fact the physical form of the highest and most complex type of associative life".
- **Conurbation** The term conurbation was coined by Patrick Geddes in 1915. This is applied to a large area of urban development that resulted from the merging of originally separate towns or cities like Greater London and Tokyo.
- **Megalopolis** Popularised by Jean Gottman (1957), this signifies super metropolitan region extending as union of conurbations, e.g. urban landscape stretching from Boston to Washington.

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- **Million City** It refers to a city whose population reaches more than one million, e.g. London reached million mark in 1800 followed by Paris in 1850 and by 1950 there were around 80 such cities.
- **Distribution of Mega Cities** The number of mega cities or megalopolis has been rising rapidly. The number of mega cities is 25 currently. At present, the number of million cities in Europe is 58, 206 in Asia, 79 in North and Central America, 43 in South America, 46 in Africa and 6 in Australia. They are also inadequate in infrastructure such as electricity sewage, disposal, health and education facilities.

Problems Of Urban Settlements

In developing countries, urban settlements suffer from unsustainable concentration of population, congested housing, lack of drinking water, poor infrastructure, no proper sewage disposal, health and education facilities, vertical expansion and growth of slums. Most of the cities in developing countries suffer from such unplanned growth. They are:

- **Economic Problems** The decreasing employment opportunities in rural areas push the unskilled and semi-skilled labour force to migrate to urban areas which is already saturated.
- **Socio-Cultural Problems** Cities in developing countries suffer from several social ills. Lack of financial resources fail to create adequate social infrastructure. Lack of employment and education tends to aggravate the crime rates. Male selective migration to the urban areas disorients the sex ratio in these cities.
- **Environmental Problems** Urban settlements in developing countries suffer from improper sewage system, massive use of fuel that causes air pollution, lack of clean drinking water, dumping of untreated wastes and huge concrete structures that aggravate the environmental problems.

Healthy City

The World Health Organisation suggests that a healthy city should have clean and safe environment, meet the basic needs of all his inhabitants, involve the community in local government and provide easily accessible health service.

Urban Strategy

The United Nations Development Programme (UNDP) outlines the urban strategy that aims to increase shelter for urban poor, provision of basic services like primary healthcare, drinking water, education, sanitation, government facilities, upgrading energy use, alternative transport system and reducing air pollution.